

3	(a)	(i)	It is the <u>sum of potential and kinetic energies</u> (of molecules/atoms/particles) <u>in random motion.</u>	1 1																				
	(a)	(ii)	There is (in ideal gas) <u>no intermolecular forces</u> so <u>no potential energy.</u> Hence, <u>internal energy is (solely) kinetic energy</u> (of particles) The (mean) <u>kinetic energy</u> (of particles) <u>proportional to (thermodynamic)</u> <u>temperature of gas.</u>	1 1 1																				
	(b)	(i)	Mean KE = $\frac{3}{2} kT = 1.5 \times 1.38 \times 10^{-23} \times (25 + 273)$ $= 6.17 \times 10^{-21} \text{ J}$	1																				
		(ii)	Using the formula, $pV = NkT$ $(6.4 \times 10^6 \text{ Pa})(1.8 \times 10^4 \times 10^{-6} \text{ m}^3) = N (1.38 \times 10^{-23} \text{ J K}^{-1}) (25 + 273.15 \text{ K})$ $N = 2.80 \times 10^{25}.$	1 1																				
		(iii)	Total internal energy = total kinetic energy of the gas $= (6.17 \times 10^{-21} \text{ J})(2.80 \times 10^{25}) = 1.73 \times 10^5 \text{ J}$	1																				
	(c)	(i)	Sections AB and CD (isothermal gradient is smaller than adiabatic) Adiabatic process will result in temperature change, i.e. the curve will be from one isothermal curve to another isothermal curve. Hence the adiabatic curve will be steeper than isothermal curves.	1																				
		(ii)	$1.6 = 160 / \text{WD}$ $\text{WD} = + 100 \text{ J}$	1																				
		(iii)	<table border="1"><thead><tr><th>Sections</th><th>Increase in internal energy / J</th><th>Work done on gas / J</th><th>Heat supplied to gas / J</th></tr></thead><tbody><tr><td>A → B</td><td>(1) 0</td><td>(7) + 100</td><td>(7) – 100</td></tr><tr><td>B → C</td><td>+ 70</td><td>(4) + 70</td><td>(2) 0</td></tr><tr><td>C → D</td><td>(1) 0</td><td>(5) – 160</td><td>(0) + 160</td></tr><tr><td>D → A</td><td>(3) – 70</td><td>(6) – 70</td><td>(2) 0</td></tr></tbody></table> Mark scheme B1 for answers (1) B1 for answers (2) B1 for answers (3) B1 for answers (4 to 6) B1 for answers (7). No mark will be awarded for (0) as it is given in the question in (ii)	Sections	Increase in internal energy / J	Work done on gas / J	Heat supplied to gas / J	A → B	(1) 0	(7) + 100	(7) – 100	B → C	+ 70	(4) + 70	(2) 0	C → D	(1) 0	(5) – 160	(0) + 160	D → A	(3) – 70	(6) – 70	(2) 0	[5]
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C → D	(1) 0	(5) – 160	(0) + 160																					
D → A	(3) – 70	(6) – 70	(2) 0																					
4	(a)		$T = 0.75 \text{ s}$																					